A Singleton Counterexample to the Toy Deep-Zero Problem

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Source problem and answer

In Problem 1.1 of [1], Hedenmalm asks the following. Given $\mathcal{E} \subset \mathbb{N}_0$, put $\mathcal{E}^c = \mathbb{N}_0 \setminus \mathcal{E}$. If f belongs to the Bargmann–Fock space and

$$f^{(j)}(0) = 0 \quad (j \in \mathcal{E}), \qquad (U_\beta f)^{(j)}(0) = 0 \quad (j \in \mathcal{E}^c),$$

must f vanish identically? Here, in the source's convention,

$$U_\alpha h(z) = e^{-\frac{1}{2}|\alpha|^2 + \bar{\alpha} z} h(z - \alpha). \tag{1}$$

The answer is *no*, already when $\mathcal{E}$ has one element.

$$|f(z)|^2 \mathrm{d}\mu_G(z) = |f(z)|^2 \mathrm{e}^{-|z|^2} \mathrm{dA}(z),$$

whose integral equals the norm of $f \in \mathscr{F}(\mathbb{C})$, the interpretation of the Fock translate becomes clearer as

$$\begin{aligned} (1.3) \quad |\mathbf{U}_\alpha f(z)|^2 \mathrm{d}\mu_G(z) &= |\mathbf{U}_\alpha f(z)|^2 \mathrm{e}^{-|z|^2} \mathrm{dA}(z) \\ &= |f(z-\alpha)|^2 \mathrm{e}^{-|z-\alpha|^2} \mathrm{dA}(z) = |f(z-\alpha)|^2 \mathrm{d}\mu_G(z-\alpha) \end{aligned}$$

means that we just perform an translation by α to the density. Consequently, the Fock translation acts isometrically on $\mathscr{F}(\mathbb{C})$,

$$\|\mathbf{U}_\alpha f\|_{\mathscr{F}(\mathbb{C})} = \|f\|_{\mathscr{F}(\mathbb{C})}, \qquad f \in \mathscr{F}(\mathbb{C}),$$

and, moreover, if $g = \mathbf{U}_\alpha f$, we may solve for f: $f = \mathbf{U}_\alpha^{-1} g = \mathbf{U}_{-\alpha} g$. This means that the isometry $\mathbf{U}_a : \mathscr{F}(\mathbb{C}) \to \mathscr{F}(\mathbb{C})$ is surjective, and hence unitary, meaning that its adjoint $\mathbf{U}_\alpha^\star$ meets $\mathbf{U}_\alpha^\star = \mathbf{U}_\alpha^{-1}$. More generally, the family of unitaries $\mathbf{U}_\alpha$ have the commutation property

$$\mathbf{U}_\alpha \mathbf{U}_\beta f = \mathrm{e}^{-\mathrm{i}\,\mathrm{Im}\,\alpha\bar{\beta}} \mathbf{U}_{\alpha+\beta} f = \mathrm{e}^{-2\mathrm{i}\,\mathrm{Im}\,\alpha\bar{\beta}} \mathbf{U}_\beta \mathbf{U}_\alpha f, \qquad f \in \mathscr{F}(\mathbb{C}),\ \alpha, \beta \in \mathbb{C},$$

which we may interpret to say that we have a projective unitary representation of the additive group $\mathbb{C}$. Let us agree to write $\mathbb{N}_0 := \{0, 1, 2, \ldots\}$. We consider the following toy problem.

Problem 1.1. ("toy problem") Let $\mathcal{E}$ be a collection of nonnegative integers, and $\mathcal{E}^c := \mathbb{N}_0 \setminus \mathcal{E}$ its complement. Suppose that $f \in \mathscr{F}(\mathbb{C})$ has the property that

$$f^{(j)}(0) = 0, \qquad j \in \mathcal{E},$$

while for some other point $\beta \in \mathbb{C}$ it holds that

$$(\mathbf{U}_\beta f)^{(j)}(0) = 0, \qquad j \in \mathcal{E}^c.$$

Does it then follow that $f = 0$?

Counterexample

Idea of the proof. Put all allowed Taylor data of $U_\beta f$ into one monomial and translate back. A suitable displacement makes the corresponding diagonal Taylor coefficient vanish. At the first level the cancellation is simply $1-|\beta|^2=0$.

Theorem 1. *Let*

$$\mathcal{E}=\{1\},\qquad \beta=1,\qquad f(z)=e^{-1/2-z}(z+1).$$

Then $0\neq f\in\mathcal{F}(\mathbb{C})$ *and*

$$f'(0)=0,\qquad (U_1f)^{(j)}(0)=0\quad \textit{for every } j\neq 1.$$

Thus Problem 1.1 has a full negative answer.

Proof. Let $g(z)=z$. Formula (1) gives

$$U_{-1}g(z)=e^{-1/2-z}g(z+1)=e^{-1/2-z}(z+1)=f(z).$$

Since g is a polynomial in $\mathcal{F}(\mathbb{C})$ and U_{-1} is unitary, $f\in\mathcal{F}(\mathbb{C})$. Moreover,

$$f'(z)=-ze^{-1/2-z},$$

so the sole condition indexed by $\mathcal{E}$ holds. The Weyl composition law has no phase for the opposite pair $1,-1$, hence

$$U_1f=U_1U_{-1}g=g=z.$$

Every derivative of z at zero except the first is zero, exactly as required for $j\in\mathcal{E}^c$. Finally, $f(0)=e^{-1/2}\neq 0$. □

The cancellation is not isolated. For $m\geq 1$ and $t=|\beta|^2$, the coefficient of z^m in $U_{-\beta}(z^m)$ is

$$e^{-t/2}P_m(t),\qquad P_m(t)=\sum_{r=0}^{m}\binom{m}{r}\frac{(-t)^r}{r!}. \tag{2}$$

Indeed, expand $e^{-\bar\beta z}(z+\beta)^m$ and collect degree m. The polynomial P_m is the Laguerre polynomial L_m. If m is odd, then $P_m(0)=1$ and its leading coefficient is $-1/m!$, so it has a positive zero t_m. Choosing $|\beta|^2=t_m$, setting $\mathcal{E}=\{m\}$, and taking $f=U_{-\beta}(z^m)$ gives the same counterexample. Theorem 1 is the case $P_1(t)=1-t$.

Scope and literature boundary

This counterexample does not conflict with the paper's Theorem 1.4, where $\mathcal{E}$ is the set of *all* even integers or all odd integers. It answers the preceding arbitrary-subset Problem 1.1 as stated.

The run's cheap indexes and a bounded current exact-title/exact-phrase search through 11 August 2026 found no prior occurrence of this singleton example. The later paper [2] proves affirmative results for Hedenmalm's distinct Problem 5.2 and congruence-class analogues via the HRT conjecture; it does not address this finite-set obstruction. The proof above is elementary and does not depend on that later work.

Verification and review recommendation

The source convention for U_α, the inverse identity, both derivative conditions, and Fock-space membership were checked directly. Formula (2) was independently re-expanded, including the factorial and phase. The $m = 1$ counterexample needs no asymptotic or external theorem. Human review is recommended, with primary attention to matching the source's sign convention for the Fock translate. Confidence is very high.

References

[1] H. Hedenmalm, *Deep zero problems*, arXiv:2205.11213 (2022), Problem 1.1 on PDF page 2, arXiv:2205.11213.

[2] Y. Li, Z. Liu, and K. Zhu, *Deep zero problems and the HRT conjecture*, arXiv:2601.09080 (2026), arXiv:2601.09080.